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**The 15th Plenary Meeting of the United Nations Global Geospatial
Information Management for Asia and the Pacific**

**The International Symposium on Geospatial Knowledge and Innovation for
Climate Resilience and Sustainable Development**

Conference Guide

21-24 September 2026, Sanya, China

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Organizers

Guided by:

United Nations Department of Economic and Social Affairs

United Nations Economic and Social Commission for Asia and the Pacific

Ministry of Natural Resources of the People's Republic of China

Organized by:

United Nations Global Geospatial Knowledge and Innovation Centre

Regional Committee of United Nations Global Geospatial Information Management for Asia and the Pacific

Hosted by:

National Geomatics Center of China

National Marine Data and Information Service

Hainan Administration of Surveying, Mapping and Geoinformation

Guangzhou Marine Geological Survey

Supported by:

Asia-Pacific Space Cooperation Organization

International Mangrove Center

Chinese Society for Geodesy, Photogrammetry and Cartography

Sanya Institute of South China Sea Geology, Guangzhou Marine Geological Survey

Sanya Yazhou Bay Science and Technology City Administration Bureau

Hainan Association of Geoinformation Industry

Hainan Society for Surveying, Mapping and Geoinformation

Hainan Institute for Marine Economic Development and Resource Conservation

State Key Laboratory of Spatial Datum

Sponsored by:

Tianditu Co., Ltd.

Beijing SuperMap Software Co., Ltd.

Feiyan Aerial Remote Sensing Technology Co., Ltd.

Guangzhou South Surveying & Mapping Technology Co., Ltd.

Cangqiong Digital Technology Co., Ltd.

Schedule Overview

21-24 September 2026 (Monday-Thursday)

Sanya Yazhou Bay Public Platform for Deep-sea Innovative Technology,
Sanya City, Hainan Province, People's Republic of China

Date	Time	Agenda	
20 Sept	12:00-20:00	Registration Day (Symposium)	
21 Sept	9:00-17:40	Session 1: United Nations Integrated Geospatial Information Framework Training Workshop	
	12:00-20:00	Registration Day (Plenary Meeting)	
22 Sept	9:00-12:30	Opening Ceremony and Keynote Speeches	
	14:00-17:30	Session 2: Land–Sea Integration for Sustainable Development	
	18:30-21:00	Welcome Reception	
23 Sept	9:00-12:30	Session 3: Leveraging GeoAI for Strengthening Land Administration, Climate Resilience and Sustainable Development	
	13:30-18:00	UN-GGIM-AP Working Group Meeting 1-5 and Executive Committee Meeting	Session 4: Youth Forum on Geospatial Knowledge and Innovation for Coastal and Marine Surveying and Monitoring
24 Sept	9:00-16:30	UN-GGIM-AP 15 th Plenary Meeting	Session 5: Strengthening National Geospatial Capacity, Policy and Sustained Cooperation
	16:30-18:30	Technical Visit	
25 Sept	All day	Departure	

Opening Ceremony

Time: 9:00-12:30, 22 September 2026

Venue: Auditorium, Building 11

Host: Mr. Pengde LI, Director, United Nations Global Geospatial Knowledge and Innovation Centre

Mr. Changsong CHEN, Director, National Geomatics Center of China

Time	Agenda	Speaker
Part A: Registration		
8:30-9:00	On-site Registration	
Part B: Opening Addresses		
9:00-9:30	Address by Ministry of Natural Resources	Mr. Bing ZHANG, Chief Planner, Ministry of Natural Resources of the People's Republic of China
	Video Address by UN Under-Secretary-General	Mr. Junhua LI, UN Under-Secretary-General for Economic and Social Affairs
	Address by Co-Chair of UN-GGIM-AP	Mr. Shailesh Kumar Sinha, Co-Chair of the United Nations Global Geospatial Information Management Asia-Pacific Regional Committee and Surveyor General of India
	Address by the People's Government of Hainan Province	Mr. Wanxin CHEN, Deputy Secretary-General, People's Government of Hainan Province
Part C: Keynote Speeches		
9:30-9:50	GeoAI Enabling Global Geospatial Public Goods Development	Mr. Jun CHEN, Academician of the Chinese Academy of Engineering
9:50-10:10	Technological Innovation Empowering High-quality Development of Natural Resources	Ms. Wenli FENG, Vice-Chair of UN-GGIM-AP, Director General of Department of Science and Technology, Ministry of Natural Resources

Time	Agenda	Speaker
10:10-10:30	Break	
10:30-10:50	The Role of GeoAI Frameworks in Ethical Governance and Global Sustainability: Standardization Considerations	Dr. Saied Pirasteh, Research Scientist Collaborator, University of Waterloo, Canada; GeoAI Expert
10:50-11:10	Geospatial Services in the AI Era	Mr. Wei HUANG, Deputy Director, National Geomatics Center of China
11:10-11:30	National Implementation of the UN-IGIF: The Republic of Korea's Experience	Dr. Woosug Cho, Co-Chair, UN-GGIM-AP
11:30-11:50	Development and Application of Marine Geospatial Information Resources in the Context of Integrated Land-Sea Management	Mr. Xingyuan REN, Deputy Director, National Marine Data and Information Service
Part D: Panel Discussion		
11:50-12:30	Theme: Innovating Geospatial Intelligence, Supporting Sustainable Development Moderator: Mr. Sandro Semedo, Geospatial Information Officer, UN-GGKIC Panelists: 1. Mr. Mehdi Samadov, Head of the International Cooperation and Public Relations Department, State Service on Property Issues under the Ministry of Economy of the Republic of Azerbaijan 2. Mr. Dheny Trie Wahyu Sampurno, Director of Atlas and Geospatial Information Utilization, Geospatial Information Agency (BIG), Indonesia 3. Mr. Suren Tovmasyan, Chairman, Cadastre Committee of the Republic of Armenia 4. Ms. Karuna Pimprasan, Geospatial Information Officer, Spatial Economy Policy Division, Geo-Informatics and Space	

Time	Agenda	Speaker
	Technology Development Agency (GISTDA) 5. Ms. María Fernanda León Pazmiño, Geographic Evaluator, Military Geographic Institute (IGM) 6. Mr. Okieno Samuels, Programme Manager, National Spatial Data Management Branch (NSDMB), Ministry of Economic Growth and Infrastructure Development, Jamaica	

Thematic Sessions

Session 1: UN-IGIF Training Workshop

Time: 09:00-17:40, 21 September 2026

Venue: #501, Building 10

Host: Ming MA, Senior Programme Manager, United Nations Global Geospatial Knowledge and Innovation Centre

Time	Agenda	Speaker
Part A: Registration		
8:30-9:00	Registration of participants.	
Part B: Welcome and Opening Remarks		
9:00-9:30	Welcoming participants, introduction, overview of the workshop objectives and expected outcomes, review of the agenda, and logistical arrangements for the day.	Ms. Ming Ma, Deputy Director, UN-GGKIC
	Opening remarks	Mr. Pengde Li, Director, UN-GGKIC
	Strengthening National Geospatial Ecosystems for a Climate-Resilient and Sustainable Future	Mr. Sandro Semedo, Geospatial Information Officer, UN-GGKIC
9:30-9:40	Group Photo	
Part C: Country Presentation		
9:40-9:52	Strengthening Indonesia's Climate Resilience through UN-IGIF Implementation	Mr. Dheny Sampurno, Director of Atlas and Geospatial Information Utilization, Geospatial Information Agency (BIG)
09:52-10:04	Integrating Geospatial Information and Digital Twins for Better Planning and Decision-Making: Singapore's Experience	Mr. Victor Khoo Hock Soon, Chief Surveyor and Director, Survey & Geomatics, Singapore Land Authority (SLA), Singapore
10:04-10:16	Strengthening National Geospatial Information Management through the	Mr. Josua Koto, Geospatial Officer, Ministry of Lands and Mineral Resources, Fiji

Time	Agenda	Speaker
	UN-IGIF: Lessons from Fiji	
10:16-10:28	UN-IGIF Implementation in Cuba: Progress, Challenges and Priorities across the Strategic Pathways	Ms. Rebeca Parada Benavente, Deputy Head of the Department of Geodesy and Cartography, National Office of Hydrography and Geodesy, Cuba
10:28-10:40	Building a Sustainable Blue Economy through Integrated Land, Coastal and Marine Geospatial Information: Kenya’s Perspective	Mr. Weldon Maritim, Director of Surveys, Survey of Kenya, Ministry of Lands, Public Works, Housing and Urban Development, Kenya
10:40-10:52	Advancements in Geospatial Information Standardization in China: Progress and Key Achievements	Ms. Jing YANG, Deputy Director, Department of Science, Technology and Standards / Department of International Cooperation, National Geomatics Centre of China
10:52-11:02	Break	
11:02-11:40	Questions, Comments and Interactions	
Part D: Plenary Discussion		
11:40-12:40	Theme: Lessons, enabling conditions and challenges in advancing country-level implementation of the UN-IGIF, with particular attention to translating the nine Strategic Pathways into sustained action and strengthening implementation support. Moderator: Mr. Sandro Semedo, Geospatial Information Officer, UN-GGKIC Panelists: 1.Mr. Enkhmanlai Anand, Director-General, Agency for Land Administration and Management, Geodesy and Cartography (ALAMGC), Mongolia 2.Mr. Hoàng Ngọc Huy, Deputy Director General, Department of Survey, Mapping and Geo-Information of Vietnam 3.Ms. María Pazmiño, Geographic Evaluator, Military Geographic Institute (IGM), Ecuador 4.Mr. Halidou Nagabila, National Geographic Institute, Burkina	

Time	Agenda	Speaker
	Faso	
12:40-14:10	Break	
Part E: Group Breakout		
14:10-17:40	Moderator: Mr. Sandro Semedo, Geospatial Information Officer, UN-GGKIC Group work on Strengthening National Geospatial Ecosystems through the Nine UN-IGIF Strategic Pathways This interactive session will enable participating Member States to examine the nine UN-IGIF Strategic Pathways as a basis for strengthening national geospatial ecosystems. Through structured parallel group work, country reflections and peer exchange, participants will consider existing national geospatial information management arrangements and capabilities, identify key strengths and implementation gaps, and define priority areas for advancing sustained implementation in accordance with national circumstances and priorities. The findings will contribute to the overall workshop outcomes and help inform possible areas for follow-up support.	

Session 2: Land-Sea Integration for Sustainable Development

Time:14:00-17:30, 22 September 2026

Venue: Auditorium, Building 11

Host: Ms. Miao Fan, National Marine Data and Information Service (NMDIS)

Time	Agenda	Speaker
14:00-14:05	Addresses	Mr. Xingyuan Ren, Deputy Director, National Marine Data and Information Service
14:05-14:25	International Mangrove Center: Strengthening Mangrove Monitoring and Scientific and Technical Cooperation for Sustainable Coastal Development	Mr. Daming BAO, Director General of the International Mangrove Center (IMC), Vice President and Secretary-General of the China Wetlands Association (CWA), and Vice Chair of the National Technical Committee on Wetland Conservation Standards
14:25-14:45	The Height System Modernization of Japan	Mr. Basara Miyahara, Director-General of Geodetic Observation Center, Geospatial Information Authority of Japan, Chair of UN-GGIM-AP Working Group 1
14:45-15:05	Overall Introduction of National Geodetic Datum of China	Mr. Qinglan Zhang, Senior Engineer, National Geomatics Center of China
15:05-15:25	Land Sea Integration in Singapore	Mr. Willis Lau, Principal Geomatics Engineer, Singapore Land Authority (SLA)
15:25-15:40	Break	
15:40-16:00	The Role of Spatial Information in Sustainable Sea-Based Development	Mr. Seyed Eskandar Seydayei, Director General of National Cartographic Center, Iran, Chair of UN-GGIM-AP Working Group 3
16:00-16:20	Study on Island Stability Induced by Global Climate Change and Resilience Enhancement	Mr. Xinran JI, Professor at Hainan University

Time	Agenda	Speaker
16:20-16:40	Integrating Land Information for Better Governance: Cabo Verde's Reforms, Achievements and Lessons Learned	Ms. Euda Helena Miranda, Executive Board Member of the National Institute for Land Management of Cabo Verde (INGT)
16:40-17:00	Satellite Remote Sensing for Integrated Land-Sea Governance in Pacific Island Countries	Ms. Wenting CAO, Associate Researcher at the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, MNR, China
17:00-17:20	Integrating CFD-PINN Hybrid Modeling and Multi-Platform Observations for Early Warning of Tropical Harmful Algal Blooms: The Hainan Pilot Study	Mr. Ming XIAO, Associate Research Fellow at the Hainan Academy of Marine Economic Development and Resource Protection
17:20-17:30	Wrap up	

**Session 3: Leveraging GeoAI for Strengthening Land
Administration, Climate Resilience and Sustainable Development**

Time: 9:00-12:30, 23 September 2026

Venue: Auditorium, Building 11

Host: Peng SHU, Senior Engineer, National Geomatics Center of China,
Director, Department of International Cooperation, Moganshan
Geospatial Information Laboratory

Time	Agenda	Speaker
Part A: Opening Remarks		
9:00-9:05	Opening Remarks	Mr. Changsong CHEN, Director, National Geomatics Center of China
Part B: Thematic Reports		
9:05-9:25	Spatial Information to Facilitate Sustainable Urban Development in Hong Kong	Prof. Xiaoli DING, Director, Research Institute for Land and Space (RILS), Chair Professor of Geomatics, Department of Land Surveying and Geospatial Science The Hong Kong Polytechnic University
9:25-9:45	Leveraging GeoAI and Earth Observation for Large-scale Biodiversity Monitoring	Prof. Tiejun WANG, Professor, School of Geospatial Engineering and Science, Sun Yat-sen University
9:45-10:00	GeoAI-Enabled Actionable Cartography for Climate Resilience and Disaster Risk Management	Prof. Jie SHEN, Professor, School of Geography, Nanjing Normal University Secretary, UN-GGIM Academic Network Executive Committee
10:00-10:15	From Geospatial Data to Climate Resilience: Sri Lanka's Experience in Integrated Geospatial Information and Disaster Response to Cyclone Ditwah	Dr. C. J. Wellage, Senior Superintendent of Surveyors, Surveyor General's Office Survey Department of Sri Lanka
10:15-10:30	From Metacognition to Meta-Action: GeoMCP and Geospatial Information Public Goods for Climate-	Dr. Mu YANG, Distinguished Expert, Institute of Global Geospatial Information and Sustainable Development,

Time	Agenda	Speaker
	Resilient Decision-Making	Moganshan Geospatial Information Laboratory
10:30-10:45	Break	
10:45-11:00	Unlocking the Value of Urban Territorial Spatial Monitoring Data for Digital and Intelligent Urban Spatial Governance	Dr. Yinxu GONG, Deputy Director, Department of Geospatial Information Analysis National Geomatics Center of China
11:00-11:15	Intelligent Interpretation of Remote Sensing Enables Sustainable Coastal Zone Management	Ms. Binru ZHAO, Engineer National Marine Data and Information Service
11:15-11:30	SAR-AI Fusion Technology: Exploring New Pathways for Intelligent Remote Sensing Monitoring	Mr. Mao ZHU, Chief Technology Officer, Beijing Vastitude Technology Co., Ltd
11:30-11:45	Geo-AI Enabled Inspection Solutions to Boost Climate Resilience Across the Asia-Pacific	Mr. Xin ZHAO, Business Development Director Guangzhou South Surveying & Mapping Technology Co., Ltd.
11:45-12:00	Geospatial Intelligence Helps Creating a Smarter World	Mr. Meng LI, General Manager of GIS Platform Product Line, SuperMap Software Co., Ltd
Part D: Roundtable Discussion		
12:00-12:30	Moderator: Mr. Shu Peng, Director, Department of International Cooperation, Moganshan Geospatial Information Laboratory; Senior Engineer, National Geomatics Center of China Speakers: 1. Dr. Victor Khoo Hock Soon, Director, Survey & Geomatics, Singapore Land Authority 2. Dr. Yang Hyo Jin, Senior Researcher, National Land Satellite Center, Ministry of Land, Infrastructure and Transport 3. Mr. Gang HAN , Deputy General Manager, GEO-Compass Information Technology Co., Ltd 4. Ms. Xiang LI, General Manager, Products Marketing Center, KQ GEO Technologies Co., Ltd 5. Prof. Jie SHEN, Professor, School of Geography, Nanjing	

Time	Agenda	Speaker
	Normal University; Secretary, UN-GGIM Academic Network Executive Committee	

Session 4: Youth Forum on Geospatial Knowledge and Innovation for Coastal and Marine Surveying and Monitoring

Time:14:00-18:00, 23 September 2026

Venue: Auditorium, Building 11

Host:

Mr. Yinan DENG, Director of the Science and Technology Division,
Guangzhou Marine Geological Survey

Mr. Hao NIU, Engineer of Sanya Institute of South China Sea Geology,
Guangzhou Marine Geological Survey

Mr. Yunqiang ZHU, Dean and professor of the School of Aerospace
Intelligence, Hainan University

Mr. Sandro Semedo, Geospatial Information Officer, United Nations
Global Geospatial Knowledge and Innovation Centre

Time	Agenda	Speaker
Part A: Opening Ceremony		
14:00-14:10	Remarks	Mr. Hanquan ZHANG, Deputy Director General, Guangzhou Marine Geological Survey
		Mr. Pengde Li, Director, UN-GGKIC
		Ms. Wenli FENG, Vice-Chair of UN-GGIM-AP, Director General of Department of Science and Technology, Ministry of Natural Resources
Part B: Keynote Reports		
14:10-14:35	Relative Sea-Level Rise in Coastal Cities: Land Subsidence Monitoring, Risk Assessment, and Adaptation Responses	Prof. Chisheng WANG, Professor, Shenzhen University
14:35-15:00	Advances and Prospects in Marine Unmanned Swarm Detection Technology	Mr. Chao DONG, Deputy Director General, South China Sea Marine Survey Center, Ministry of Natural Resources , China
Part C: Youth Sharing Session		

Time	Agenda	Speaker
15:00-15:10	Height Datum Transfer and Unification for Islands and Reefs	Dr. Baogui KE, Researcher, Chinese Academy of Surveying and Mapping
15:10-15:20	Tidal Flat Topography Inversion from Multi-source Satellite Altimetry and Remote Sensing Imagery	Assoc, Prof. Nan XU, Shenzhen University
15:20-15:30	Exploration and Application of Tropical Remote Sensing Intelligent Interpretation Technology: A Case Study of Hainan Island	Dr. Chunxiao WANG, Professor of Engineering, Hainan Geomatics Center, Ministry of Natural Resources, China
15:30-15:40	Refined Tropospheric ZWD Model for the Tibetan Plateau Under Oceanic Moisture Influence	Dr. Chen LIU, Postdoctoral Researcher, National Geomatics Center of China
15:40-15:50	Integrated RS and GIS in Coastal Eco-Geological Surveying: Insights from Representative Case Studies	Mr. Yi DENG, Engineer, Guangzhou Marine Geological Survey
15:50-16:05	Break	
16:05-16:15	Integrated Space-Air-Ground-Sea Digital Twin System: Empowering Coastal Resilience Building and Smart Ocean Monitoring	Mr. Ziyang ZHAO, General Manager Feiyan Aerial Remote Sensing Tech Co., Ltd.
16:15-16:25	Practice of Land-Sea Integrated 3D Visualization Technology and Reflections on Its Application Potential for Island Countries	Mr. Xiang TANG, Senior Engineer, Sanya Institute of South China Sea Geology, Guangzhou Marine Geological Survey
16:25-16:35	Integrating Land, Coastal and Marine Geospatial Information: Indonesia's Experience	Mr. Dheny Trie Wahyu Sampurno, Director of Atlas and Geospatial Information Utilization, Geospatial Information Agency (BIG), Indonesia

Time	Agenda	Speaker
16:35-16:45	Building Coastal Resilience through Integrated Geospatial Information: Jamaica's Experience	Mr. Okieno Samuels, Project Manager, National Spatial Data Management Branch (NSDMB), Ministry of Economic, Growth and Infrastructure Development, Jamaica
16:45-16:55	Natural Resources Intelligent Monitoring and Knowledge Services	Mr. Huaqiao XING, Vice Dean, Shandong Jianzhu University
16:55-17:05	Automated Geo-computations Driven by Knowledge Graphs	Dr. Yanmin QI, Postdoctor, Hainan University
Part D: Roundtable Discussion		
17:05-18:00	Theme: New Pathways for the Integration and Collaborative Monitoring of Marine Surveying and Mapping Technologies across Land, Sea, Air, and Space Moderator: Prof. Yunqiang ZHU, Dean, School of Aerospace Intelligence, Hainan University Speakers: 1. Prof. Chisheng WANG, Professor, Shenzhen University 2. Mr. Ziyang ZHAO, General Manager, Feiyan Aerial Remote Sensing Tech Co., Ltd 3. Mr. Sandro Semedo, Geospatial Information Officer, UN-GGKIC 4. Dr. Yutong FU, Deputy Director, Coastal Zone Comprehensive Survey Division, Sanya Institute of South China Sea Geology, Guangzhou Marine Geological Survey	

Session 5: Strengthening National Geospatial Capacity, Policy and Sustained Cooperation

Time: 9:00 – 16:30 September 24, 2026

Venue: Building 10, Room 522, Deep Sea Science and Technology Innovation Public Platform

Host: Sandro Semedo, Geospatial Information Officer, UN-GGKIC

Time	Agenda	Speaker
Part A: Expert Presentation		
9:00-9:15	Deepening International Cooperation on Geospatial Information for Joint Climate Resilience and Sustainable Development in the Asia-Pacific	Dr. Hui JIANG, Secretary General, Asia Pacific Space Cooperation Organization (APSCO)
9:15-9:30	Exploration of S-100 based Nautical Chart Production, Quality Verification, and Application	Ms. Shujuan JIA, Senior Engineer, Shanghai Nautical Chart Center, China Maritime Safety Administration
9:30-9:45	Delivering National Geospatial Information Services at Scale: The Tianditu National Platform for Common Geospatial Information Services	Mr. Hongping ZHANG, Senior Engineer, National Geomatics Center of China
9:45-10:00	Establishing a Common Geospatial Reference Framework for Integrated Land, Coastal and Ocean Information: Papua New Guinea's Experience	Mr. Nidkombu Edwin, Assistant Surveyor General, Papua New Guinea
10:00-10:15	Society with advanced use of geospatial information in Japan and Fundamentals for the achievement of advanced utilization of geospatial information,	Hayato FUKUOKA, PASCO CORPORATION
10:15-10:30	Break	

Time	Agenda	Speaker
Part B: Panel Discussion		
10:30-11:15	Theme: Advancing Regional Cooperation for Coastal and Ocean Geospatial Information Moderator: Mr. Okieno Samuels, Programme Manager, National Spatial Data Management Branch (NSDMB), Ministry of Economic Growth and Infrastructure Development, Jamaica Panelists: 1. Mr. Weldon Maritim, Director of Surveys, Survey of Kenya, Ministry of Lands, Public Works, Housing and Urban Development, Kenya 2. Dr. Hui JIANG, Secretary General, Asia Pacific Space Cooperation Organization (APSCO) 3. Hayato FUKUOKA, PASCO CORPORATION	
Part C: Outcomes		
11:15-11:30	Summary of Achievements	
12:00-13:30	Break	
Part D: Breakout Discussions		
14:00-14:30	Where can regional cooperation add value? Participants exchange experiences and perspectives on areas where regional and subregional cooperation could support stronger geospatial information management.	
Part E: Report-back and Regional Dialogue		
14:30-15:30	A designated rapporteur or spokesperson from each group will share key observations and considerations emerging from the discussions. The moderated dialogue will provide an opportunity to reflect on common themes, different regional perspectives and areas that may merit further exchange.	
15:30-15:45	Synthesis of the main findings, priorities and considerations emerging from the four-day programme.	

UN-GGIM-AP Working Group Meetings (By Invitation Only)

Time: 13:30–16:00, 23 September 2026

Venue:

#520, Building 10

UN-GGIM-AP Working Group 1 Meeting

#412, Building 10

UN-GGIM-AP Working Group 2 Meeting

#314, Building 10

UN-GGIM-AP Working Group 3 Meeting

#313, Building 10

UN-GGIM-AP Working Group 4 Meeting

#311, Building 10

UN-GGIM-AP Working Group 5 Meeting

Themes:

WG1: Geodetic Reference Frame

WG2: Integrated Geospatial Information for Effective Land
Administration

WG3: Integrating Geospatial Information and Statistics

WG4: Integrated Geospatial Information Framework

WG5: GeoAI for Sustainable Development

**UN-GGIM-AP Executive Board Meeting
(By Invitation Only)**

Time: 16:30–17:30, 23 September 2026

Venue: #311, Building 10

Agenda:

Preparations for the next EB Meeting

Preparations for the 16th Plenary Meeting in 2027

**UN-GGIM-AP 15th Plenary Meeting
(By Invitation Only)**

Time: 9:00–16:30, 24 September 2026

Venue: #311, Building 10

Time	Agenda
Part A: UN-GGIM-AP Plenary Opening	
9:00-9:05	Welcome to the Plenary Meeting by the host country
9:05-9:20	Co-Chairs of UN-GGIM-AP
9:20-9:25	Welcome to the Plenary by UN-GGIM Secretariat
Part B: UN-GGIM-AP Plenary Session I	
9:25-9:30	Agenda item 1: Confirmation of Agenda
9:30-9:40	Agenda item 2: Executive report of UN-GGIM-AP
9:40-9:50	Agenda item 3: Liaison Reports from UNSD (Secretariat of UN-GGIM)
9:50-10:00	Agenda item 4: Report from UNESCAP
10:00-10:20	Break
10:20-11:10	Agenda item 5: Reports from the UN-GGIM-AP Working Groups
11:10-11:20	Agenda item 6: Reports from UN-GGIM Regional Committees
11:20-11:30	Agenda item 7: Reports from UN-GGIM-AP Networks
11:30-11:40	Agenda item 8: Regional, Interregional, and Global Cooperation
11:40-12:00	Comments and Discussion
12:00-13:30	Lunch Break
Part C: UN-GGIM-AP Plenary Session II	
13:30-14:00	Agenda item 9: Reports on the Working Groups' Plan (2027 - 2028), by Chair of each WG
14:00-14:30	Agenda item 10: Discussion on the next Plenary meeting

Time	Agenda
14:30-16:00	Agenda item 11: Discussion and adoption of the Resolution
Part D: UN-GGIM-AP Plenary Closing	
16:00-16:10	Agenda item 12: Handover the UN-GGIM-AP Flag to the next host
16:10-16:30	Agenda item 13: Remarks by the UN-GGIM-AP Co-Chairs and host country

Technical Visit

Time: 16:30-18:00, 24 September 2026

Venue:

Sanya Institute of South China Sea Geology, Guangzhou Marine Geological Survey

The visit includes two parts: first, business display boards covering marine surveying and mapping, geographic information product services, coastline ecological restoration, and the development and utilization of new marine energy resources; second, research on marine survey equipment developed under the "Eight Dragons Exploring the Sea" program. The visit will last approximately 40 minutes.

Yazhou Bay Science City Industrial Promotion Achievement Exhibition Center

Tour the center's modules, including the Preface Hall, Land-Nanfan Exhibition Hall, Marine-Deep-Sea Exhibition Hall, and Intellectual Property Exhibition Hall. The visit will last approximately 35 minutes.

Welcome Reception

(Business Casual is fine)

Time: 18:30–21:00, 22 September 2026

Venue: Yazhou Hall, Curio Collection by Hilton, Sanya Yazhou Bay Qianhe International Exchange Center

Attendance Information



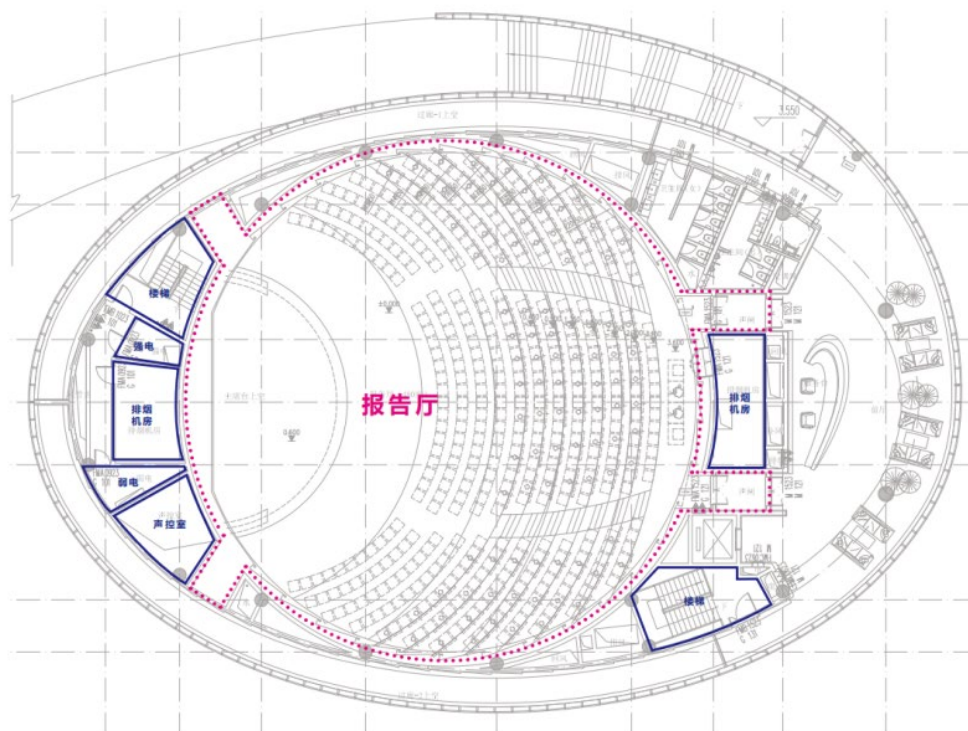
Venue: Sanya Yazhou Bay Public Platform for Deep-sea Innovative Technology

Address: Yazhou District, Sanya City, Hainan Province, People's Republic of China



Auditorium, Building 11

Opening Ceremony, Keynote Speeches, Session 2, Session 3, Session 4



#501, Building 10

Session 1, UN-GGIM-AP 15th Plenary Meeting

#522, Building 10

Session 5

#520, Building 10

UN-GGIM-AP Working Group 1 Meeting

#412, Building 10

UN-GGIM-AP Working Group 2 Meeting

#314, Building 10

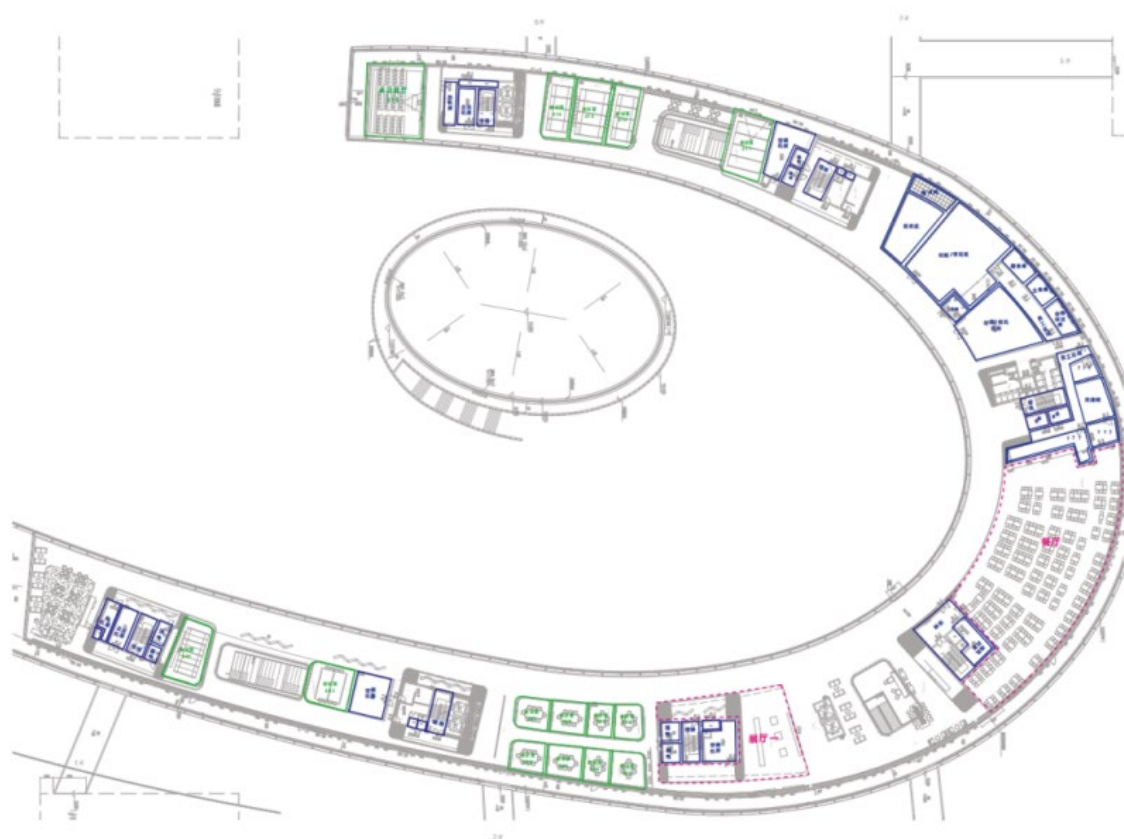
UN-GGIM-AP Working Group 3 Meeting

#313, Building 10

UN-GGIM-AP Working Group 4 Meeting

#311, Building 10

UN-GGIM-AP Working Group 5 Meeting and Executive Board Meeting



Hotels

Crowne Plaza Sanya Yazhou Bay

Address:

No.1 Xindao Street, Yazhou Bay Science and Technology City,
Sanya, Hainan Province, China

Contact:+86-898-3826-9999

Curio Collection by Hilton Sanya Yazhou Bay Qianhe International Exchange Center

Address:

Chuangyi Xincheng, Yazhou Bay, Yazhou District, Sanya City,
Hainan Province, China

Contact:+86-898-3827-3999

Overseas Currency Exchange Service

Note: Service hours 9:00 – 17:00, from Monday to Sunday. Some
currencies require advance reservation.

Bank of China, Sanya Yazhou Sub-branch

Address: Room 1011–1012, 1st Floor, Building 4, Baitai Industrial
Park, Yazhou Science and Technology City, Yazhou District, Sanya
City, Hainan Province, China.

China Construction Bank, Sanya Yazhou Sub-branch

Address: Shop No. 3–5, Building D, Business Building,
Comprehensive Service Center, Yazhou Bay Free Trade Center,
Intersection of Zhenzhou Road and Chuangyi Road, Yazhou Bay
Science and Technology City, Yazhou District, Sanya City, Hainan
Province, China

Transportation Services

1. Shuttle Bus

Shuttle bus services will operate between the venue and two hotels: Crowne Plaza Sanya Yazhou Bay and Hilton Garden Inn Sanya Yazhou Bay Qianhe International Exchange Center, from 21 to 24 September.

Service hours: 8:00-8:45, 12:00-13:45, 17:30-18:30 daily.

A group shuttle for the technical site visit on 24 September will be available from 16:30 to 18:30.

2. Transport Hubs

Airport: Sanya Phoenix International Airport is approximately 35 kilometres from the venue.

Bus Route 409 (Phoenix Airport Station-Yongyou Industrial Park Station) takes about 1 hour and 20 minutes; a taxi ride takes around 40 minutes.

High-speed Railway Station: Yazhou Station is approximately 7 kilometres from the venue.

Bus Route 306 (Yazhou Railway Station-Puhai Community Station) takes about 1 hour and 10 minutes; a taxi ride takes around 18 minutes.

Medical Facility



Sanya Yazhou Bay Science City Hospital

Address: No. 9 Huanjin Road, Yazhou District, Sanya City, Hainan Province

24-hour Emergency Nurse Station Tel: +86 898 38880120

In case of emergency, please call 120 first.

Catering

Shopping Mall with Gourmet Dining	
Zhaoshang Garden	No. 16 Yumin Road, Yazhou District
Chinese Restaurant	
Jiuru Ge · Chinese Restaurant	Room A101, Building 1, Zhaoshang Deep Sea Equipment Industrial Park, Zhenzhou Road, Yazhou District
Qinanshan · Private Kitchen	No. 6, Nanshan 2nd Group, Nanshan Village Committee, Yazhou District
Wenjian · Chinese Hainan Cuisine · Seaview Restaurant (Yazhou Branch)	23rd Floor, Crowne Plaza Sanya Yazhou Bay, No. 1 Xindao Street, Yazhou District
International Cuisine Restaurant	
Yue Restaurant · Western · Asian Flavors (Crowne Plaza Sanya Yazhou Bay)	Crowne Plaza Sanya Yazhou Bay, Building 1, No. 1 Xindao Street, Science and Technology City, Yazhou District
Haoyueke Steakhouse (Sanya Yazhou Zhaoshang Garden Branch)	Shop L4003, 4th Floor, Zhaoshang Garden, Chuangyi Road, Yazhou District
Halal Food	
Jingmulan Lanzhou Beef Noodles · Halal (Yazhou Branch)	900 Meters West of the Intersection of Huanjin Road and G225, South Side of the Road, Yazhou District
Halal Lanzhou Beef Hand-Pulled Noodles (Poly · Qianhai Branch)	Room 105, Building 5, Poly Qianhai Commercial Shops, Yazhou District
Fast Food	
Pizza Hut (Sanya Zhaoshang Garden Branch)	1st Floor, Gate 2, Zhaoshang Garden, Intersection of Yumin Road and Chuangyi Road, Yazhou District (Right Side of Parking Entrance)
McDonald's (Yazhou Zhaoshang Garden Branch)	L1026, Zhaoshang Garden, Comprehensive Service Center of Deep Sea Science and Technology City, Yazhou District

KFC (Yazhou Science and Technology City Branch)	Shop 1, 1st Floor, Deep Sea Cube Cultural Industry Park, Chuangxin Road, Yazhou Bay Science and Technology City, Yazhou District
Cafe& Tea & Bar	
Cascade Bar (Crowne Plaza Sanya Yazhou Bay)	Crowne Plaza Sanya Yazhou Bay, Building 1, No. 1 Xindao Street, Science and Technology City, Yazhou Town, Yazhou District
Buer Sunset Music Bistro	Yazhou Bay Beach, near Crowne Plaza
Starbucks Coffee	No. 16 Yumin Road, Yazhou District
Luckin Coffee	Rooms 101-102, Building B, Research Building, Comprehensive Service Center Project of Deep Sea Science and Technology City, Zhenzhou Avenue, Yazhou District
Cotti Coffee (Yazhouwan University Town Store)	Room 104, 1st Floor, Innovation Research and Study Valley Project, Yazhou Bay Science and Technology City, Chuangxin Road, Yazhou District
Nanshan Temple Dongli · Coffee · Tea	North Side of Muyin Qinghuan Du Mountain View Homestay, Yazhou District
Local Cuisine	
Chuangwei · Hainan Folk Cuisine · Intangible Cultural Heritage Tengqiao Pork Ribs	Building 4, Baitai Bio-Tech Park, Science and Technology City, Yazhou District
Lixiang · Hainan Li Ethnic Cuisine	No. 74, Nanshan 2nd Group, Nanshan Village Committee, Yazhou District
Along Seafood Farmhouse · Sanya Time-Honored Brand (Nanshan Seaside Branch)	Sima Village, Yazhou District
Qiongzhou · Zaopo Vinegar Hot Pot (Yazhou Branch)	Xinghai Night Market Commercial Street, Intersection of Xindao Street and Chuangxin Road, Yazhou Science and Technology City, Yazhou District (Near Crowne Plaza and Yazhou Bay Seaside)
Yunfan Tea Market · Hainan Laoba Tea (Yazhou Science	Rooms 103-106, Building 6, Area B, Phase II, Hangtian Yunhaitai, Yazhou

and Technology City Branch)

Bay Science and Technology City,
Yazhou Town, Yazhou District

Speaker Information

Jun CHEN

Presentation Title

Speaker's BIO

Presentation Outline



Wenli FENG

Presentation Title

Technological Innovation Empowering High-quality Development of Natural Resources

Speaker's BIO

Dr. Wenli FENG is a research fellow. She is mainly engaged in research on the implementation of major projects, technical systems, and policies in the fields of natural resources survey and monitoring, real estate registration, and territorial spatial planning. She currently serves as Director-General of the Department of Science and Technology Development of the Ministry of Natural Resources of China, Chairman of the National Technical Committee on Standardization of Natural Resources and Territorial Spatial Planning (TC93), and Vice Chair of the United Nations Committee of Experts on Global Geospatial Information Management of the Asia Pacific Regional Committee (UNGGIM-AP).

Presentation Outline

In the face of global challenges such as climate change and ecological security, China has made scientific and technological innovation the core engine for the high-quality development of natural resources, and achieved significant breakthroughs in intelligent spatial dynamic monitoring, deep-sea equipment, satellite remote sensing, and ecological governance. Going forward, China will adhere to green, digital-intelligent, collaborative, and international approaches, empowering the modern governance of territorial space with science and technology, enhancing the capacity for early warning and prevention of ecological security risks, and building a broad framework for scientific and technological innovation that spans from mountaintops to oceans, thereby contributing to the United Nations 2030 Sustainable Development Goals.



Saied Pirasteh

Presentation Title

The Role of GeoAI Frameworks in Ethics Governance and Global Sustainability: Standardization Considerations

Speaker's BIO

Dr. Saied Pirasteh is Professor and Associate Dean at the Institute of Artificial Intelligence, Shaoxing University, China, specializing in GeoAI, environmental disasters, and sustainability. He is also a Visiting Professor at the University of Waterloo and a Distinguished Professor at the University of South Africa. He is an Academician of the Asian Academy of Sciences, a Fellow of the IET, and an IEEE Senior Member. He is the inventor and co-founder of the Geospatial Infrastructure Management Ecosystem (GeoIME), with over 300 publications and 9,300 citations. He holds key roles in international platforms including the UN-GGIM Academic Network, the ISPRS Disaster Management Working Group, the CIG Overseas Network, and the IEEE GRSS Disaster Management and Early Warning Committee. His research advances GeoAI Plus, GeoIME-GPT/GeoGPT, Agentic AI, and geospatial cognitive ecosystems for climate resilience and disaster risk reduction.

Presentation Outline

This presentation examines the evolution of GeoAI from an emerging technological capability toward a trusted foundation for global decision intelligence and sustainable development. It first establishes GeoAI's role in integrating artificial intelligence, geospatial data, and spatial reasoning, illustrated through an example of the GeoIME and GeoIME-GPT frameworks for infrastructure vulnerability and disaster-risk assessment. It then examines critical gaps in GeoAI-specific standards, including

interoperability, reproducibility, data quality, privacy, bias, transparency, accountability, and cybersecurity. The presentation proposes an integrated, lifecycle-based Responsible GeoAI framework connecting AI standards, geospatial standards, ethical principles, and sustainability objectives. Particular attention is given to multi-stakeholder governance, AI procurement, geolocation regulation, education, and practical organizational actions. Finally, it explores emerging frontiers, including real-time geospatial decision intelligence and promptless GeoAI—and argues that as GeoAI becomes increasingly autonomous, trust, governance, and standardization must evolve alongside technological innovation.



Wei HUANG

Presentation Title

Geospatial Services in the AI Era

Speaker's BIO

Mr. Wei HUANG specializes in theoretical research, technology development, engineering implementation, and organizational management in the field of geographic information applications and services. He has presided over, participated in, or planned and advanced more than 20 national and provincial/ministerial-level geospatial information engineering or research projects, and led the construction of the national geospatial information public service platform (Tianditu), which serves as China's national geospatial infrastructure.

Presentation Outline

This report takes the National Geographic Information Public Service Platform "Tianditu" as its point of departure, systematically reviewing the achievements and major challenges of China's geographic information public services. Driven by the dual forces of artificial intelligence and digital transformation, it focuses on the new demands placed on geographic information services by data as a new factor of production. The report explores pathways for building new models of geographic information services in the AI era, and offers forward-looking systemic reflections and targeted recommendations across dimensions including technical architecture optimization, dynamic data updating, and product-service innovation.

Woosug Cho



Xingyuan REN

Presentation Title

Development and Applications of Marine Geospatial Information Resources under Land-Sea Integration

Speaker's BIO

Xingyuan REN is a Professor-level Senior Engineer and Deputy Director of the National Marine Data and Information Service. He has served as project leader or technical lead for several national, provincial, or ministerial-level projects, including the Smart Ocean Emergency Communication Pilot Network, construction of the Zhejiang Provincial Big Data Center, and Antarctic geospatial information projects. He has led major work on the development of a unified land-sea vertical datum, intelligent interpretation of coastal-zone remote sensing imagery, intelligent compilation of marine thematic maps, digital representation of typical deep-sea habitats, naming of undersea features, and international cooperation in marine geospatial information. A project he led on key technologies for marine emergency communication networks received the First Prize of the Marine Engineering Science and Technology Award. Over the past five years, he has published more than 10 papers, authored one monograph, been granted six national invention patents, and drafted eight standards.

Presentation Outline

This presentation focuses on the development and applications of marine geospatial information resources under land-sea integration. It highlights

the significance of land-sea integration for global ocean governance, territorial spatial planning, marine economic development, and disaster prevention and mitigation, and analyzes the major challenges in this field. Based on these challenges, China's practices in marine geospatial information resource development are introduced through key technologies, including three-dimensional marine geospatial data acquisition, land-sea vertical datum unification, and multi-source geospatial data integration. Various marine geospatial information products and related application services are also presented.



Dheny Trie Wahyu Sampurno

Presentation Title:

Strengthening Indonesia's Climate Resilience through UN-IGIF Implementation.

Speaker's BIO

Dheny graduated from Kyoto University, majoring in Environmental Science in 2020. He holds a background in geography and has been engaged in geospatial-related management work in Indonesia since 2003. His research interests cover remote sensing, surveying, data modelling, geospatial ecosystems and the geospatial industry, and he possesses solid professional competence in geospatial management.

Presentation Outline

Indonesia, as the world's largest archipelagic country, faces an existential threat from climate change, with potential economic losses in key sectors estimated to reach IDR 115.4 trillion by 2024. While the government has integrated climate resilience into its national priorities, current efforts are hampered by institutional fragmentation, data silos, and a critical "scale gap" in geospatial information. To transition from broad national targets to effective local interventions, Indonesia must adopt the United Nations Integrated Geospatial Information Framework (UN-IGIF). This framework provides the strategic pathways—spanning governance, technology, and people—necessary to build a GeoClimate Data Hub and a high-resolution (1:5,000) spatial foundation for mitigation, adaptation,

and monitoring.



Victor Khoo Hock Soon

Presentation Title:

Towards SG Digital Twin - a digital journey supported by Geospatial Ecosystem

Speaker's BIO

Dr. Victor Khoo is the Chief Surveyor and Director of Survey and Geomatics at the Singapore Land Authority (SLA), where he leads Singapore's national mapping programmes and geospatial strategies. As Chief Surveyor, he oversees the national cadastral survey system, positioning infrastructure, and the National 3D Mapping Programme. Dr. Khoo serves as President of the Land Surveyors Board Singapore. Internationally, he is a member of the International Advisory Committee of the UN Global Geodetic Centre of Excellence (UN-GGCE) and Vice-Chair of UN-GGIM Asia-Pacific, contributing to the advancement of geodesy and geospatial information management in the region. He is also an Adjunct Professor at Nanyang Technological University (NTU), contributing his professional experience to education and capability development in the geospatial field.

Presentation Outline

Singapore's path to a national urban digital twin rests on a robust, integrated geospatial ecosystem that turns authoritative data into trusted decisions. This session outlines that journey - from the SLA's Whole-of-Government National 3D Mapping Programme, which combines aerial and nationwide mobile mapping to deliver high-resolution terrain models, CityGML LOD2 building models, point clouds and reality meshes, to the applications they enable: flood risk assessment and climate resilience, urban planning, sustainability and solar analysis, and emergency response. It also covers real-time precise positioning through SiReNT for

smart mobility, GeoAI for automated map production, and drone-based high-frequency updates, closing with SLA's role as national mapping and geospatial agency underpinning a safe, sustainable and smart city.

Josua Koto

Presentation Title

Strengthening National Geospatial Information Management through the UN-IGIF: Lessons from Fiji

Speaker's BIO

GIS and geospatial professional focused on modernizing land administration through spatial data, cadastral GIS, and geospatial technology. My work combines cadastral mapping, land information management, spatial analysis, photogrammetry, drone surveying, and web GIS to improve how spatial information is captured, managed, analyzed, and shared.

Experience in working with ArcGIS Pro, ArcGIS Enterprise and Online, RTK surveying, UAV mapping, remote sensing, and spatial databases. I am particularly interested in transforming traditional land and mapping workflows into integrated, digital geospatial systems that support better decision-making and public-sector service delivery.

Presentation Outline

Governance

The current situation of Governance and Institutions, Policy and Legal and Finance. The way forward (Case Study: The FGIMC Council)

Technology

The current situation of Data, Innovation and Standards in Fiji the direction The way forward (Case Study: VanuaHub)

People

The current situation of Partnership, Capacity & Education and Communication and Engagement. The way forward (Case Study: Forest Cover Working Group)

Rebeca Parada Benavente

Presentation Title

Implementation of the UN-IGIF for Decision-Making in Cuba, Future Challenges

Speaker's BIO

A 2003 graduate with a degree in Hydrography and Geodesy Engineering, I have extensive experience in Geographic Information Systems, multiscale cartography, and database management. My work includes the creation of the Spatial Data Infrastructure of the Republic of Cuba, the development of a website for the National Office of Hydrography and Geodesy, the creation and development of a Geographic Information System for the Provincial Offices of Hydrography and Geodesy, the creation of the Land Coverage Map of the Republic of Cuba using the mad-mex tool, and the creation and development of a Geographic Information System for the Intermediate Care Unit (UCIM) of the "Dr. Eduardo Agramonte Piña" Pediatric Hospital in Camagüey, among other projects.

I currently serve as the Deputy Head of the Geodesy and Cartography Department at the National Office of Hydrography and Geodesy. I am also the Chair of the Technical Subcommittee on Cartography and Mapping, which establishes and approves Cuban standards for project implementation, taking into account international standards to ensure data quality. For the past two years, I have been developing a doctoral thesis on the implementation of the UN-IGIF Integrated Geospatial Information Framework. The goal is to create a standardized model to enable entities that generate geospatial information to work uniformly and guarantee the interoperability of geospatial data, thereby contributing to the indicators proposed in the Sustainable Development Goals.

Presentation Outline

This presentation addresses the modernization of Cuba's Spatial Data Infrastructure (IDERC) through the implementation of the United Nations Integrated Geospatial Information Framework (UN-IGIF).

Organized into 12 strategic slides, the exhibition synthesizes the historical evolution of Cuban cartography, the consolidation of the National Geodetic Network and the current state of topographic coverage.

It also delves into the three pillars of the framework (Governance, Technology, and Society) and its nine fundamental principles. Finally, it presents key projects with socioeconomic impact in precision agriculture, disaster risk management, and food sovereignty, concluding with the national action plan aligned with the Sustainable Development Goals (SDGs) and the priority targets for 2030.

1. Introduction and Context General framework of the UN-IGIF and the historical development of cartography in Cuba.
2. Basic Infrastructure National geodetic network, reference systems and current topographic coverage.
3. Pillars and Principles Application of Governance, Technology and Society according to the 9 UN-IGIF principles.
4. Strategic Projects Case studies: food sovereignty, risk management and precision agriculture.
5. Conclusions and Future Roadmap, alignment with the SDGs and strategic goals towards 2030.

Weldon Maritim

Presentation Title

Building a Sustainable Blue Economy through Integrated Land, Coastal
and Marine Geospatial Information: Kenya's Perspective

Speaker's BIO

Presentation Outline



Jing YANG

Presentation Title:

Advancements in Geospatial Information Standardization in China:
Progress and Key Achievements

Speaker's BIO

Ms. Jing YANG serves as Deputy Director of the Department of Science, Technology and Standards (Department of International Cooperation) at NGCC. She has long served as the National Mirror Committee of ISO/TC 211 in China, and is dedicated to research on geographic information standardization both at home and abroad.

Presentation Outline

Representing the National Mirror Committee of ISO/TC 211 in China, this presentation outlines the progress and core achievements in China's geospatial information standardization, with a particular focus on the internationalization of standards. Building on these advancements, it concludes by addressing current challenges in talent cultivation and international collaboration while offering a vision for a smarter, standardized future.



Daming BAO

Presentation Title:

International Mangrove Center: Strengthening Mangrove Monitoring and Scientific and Technical Cooperation for Sustainable Coastal Development

Speaker's BIO

Prof. Daming BAO, a professor-level senior engineer, currently serves as Director General of the International Mangrove Center (IMC). He is also Vice President and Secretary-General of the China Wetlands Association (CWA), and Vice Chair of the National Technical Committee on Wetland Conservation Standards. He previously served as the Executive Director General of the Department of Wetland Management, National Forestry and Grassland Administration (NFGA) of China. Prof. Bao has dedicated his career to advancing wetland conservation policy, planning, and international cooperation. He led the formulation and implementation of major national policies and initiatives, including the Wetland Conservation Law of the People's Republic of China, the Special Action Plan for Mangrove Conservation and Restoration (2020–2025), and the National Wetland Conservation Plan (2022–2030). He oversaw two national wetland inventories, supported the establishment of China's wetland monitoring and assessment system, and played an active role in advancing China's engagement in the Convention on Wetlands and multilateral cooperation.

Presentation Outline

The presentation will outline key challenges to mangrove conservation worldwide and provide an overview of the International Mangrove Center (IMC). It will highlight the IMC's work in mangrove monitoring, knowledge sharing and capacity-building, and explain how geospatial information supports mangrove resource inventories, conservation and restoration. It will also identify opportunities for future cooperation in data sharing, technical cooperation and regional collaboration to help island and coastal countries strengthen mangrove conservation and integrated land–coast–ocean management.

Basara Miyahara

Presentation Title

The Height System Modernization

Speaker's BIO

Presentation Outline



Qinglan ZHANG

Presentation Title

Overall Introduction of National Geodetic Datum of China

Speaker's BIO

Mr. Qinglan ZHANG, Senior Engineer of National Geomatics Center of China, was selected as a young scientific and technological talent by the Ministry of Natural Resources. He has been engaged in the construction, maintenance, and service of national geodetic datum of China for a long time, and has participated in China national geodetic datum modernization construction.

Presentation Outline

The construction and development of China national modern geodetic datum include infrastructure construction and application services such as the National Satellite Navigation and Positioning Station Network, National GNSS Geodetic Control Network, National vertical datum Network, National Gravity Reference Network, and National Geodetic Datum Service System.

**Willis Lau****Presentation Title**

Land Sea Integration in Singapore

Speaker's BIO

Willis Lau is a Principal Geomatics Engineer at the Singapore Land Authority (SLA), working across both the Geodesy and Mapping teams. In his two years at SLA, Willis has been contributing to the establishment of Singapore's land motion monitoring system, facilitating researcher access to SLA's mapping data, and supporting Singapore's chairing of Working Group 2 of the UN-GGIM-AP, contributing to the advancement of geodesy and geospatial information management in the region.

Presentation Outline

Climate change poses significant challenges for the Asia-Pacific region, with rising sea levels compounded by land subsidence emerging as pressing concerns. Addressing these threats requires land-sea integration to ensure that terrestrial and maritime data are captured within a unified framework and can be meaningfully analysed together.

A key challenge in achieving this lies in bridging the gap between land and sea domains, which this presentation explores through the application of interpolation methods to integrate data across both environments. This presentation outlines Singapore's efforts in land-sea integration, highlighting both existing work and the road ahead as Singapore works towards land sea integration for enhanced climate resilience.



Seyed Eskandar Seydayei

Presentation Title

The Role of Spatial Information in Sustainable Sea-Based Development

Speaker's BIO

Mr. Seyed Eskandar Seydayei is the Director General of the National Cartographic Center (NCC) of the Islamic Republic of Iran, the country's principal authority for surveying, mapping, and geospatial information. He serves as Chair of UN-GGIM-AP Working Group 3: Integrating Geospatial Information and Statistics, a role in which he has been reappointed in recognition of Iran's commitment and expertise in integrating statistical and geospatial data. Under his leadership, WG3 promotes the use of standards for geospatial-statistical integration, strengthens regional cooperation, and contributes to the Sustainable Development Goals.

Mr. Seydayei has been a prominent advocate for Iran's active engagement in global scientific forums and has overseen NCC initiatives in hydrographic mapping, mineral resource assessment, and national spatial data infrastructure development. He represents Iran on the UN-GGIM-AP Executive Board and regularly contributes to international geospatial governance discussions.

Presentation Outline

This presentation by Iran's National Cartographic Center (NCC) examines how spatial information supports sustainable sea-based development. It highlights land-sea interactions and the need to integrate terrestrial and marine geospatial data for planning, resource management, environmental protection, and conflict resolution. The NCC's role includes reference frames, positioning services, mapping, hydrography,

nautical charts, and national spatial data infrastructure. Its Marine SDI contains more than ten thousand layers, complemented by coastal monitoring stations, real-time sea-level services, tidal modelling, and extensive nautical charting in the Persian Gulf, Caspian Sea, and Oman Sea. Challenges include data compatibility, availability, dynamic environments, and stakeholder cooperation. Future plans focus on coastal mapping, hydrography, offshore positioning, and MSDI development. The core message: “One sea, one picture, better decisions.”



Xinran JI

Presentation Title

Island Stability under Global Climate Change and Approaches to Enhancing Resilience

Speaker's BIO

Dr. Xinran JI is a Professor at Hainan University, Invited Expert of Blue Eco Geomatics (Hainan Province) Co., Ltd., and a Leading-level Talent of Hainan Province. His research focuses on hydrodynamic characteristics and ecological security protection of South China Sea islands and reefs. He has led one project under the National Key R&D Program of China, three projects funded by the National Natural Science Foundation of China, and four provincial-ministerial-level research projects. His awards include the First Prize for Scientific and Technological Progress from the Chinese Society of Naval Architecture and Marine Engineering, the First Prize for Scientific and Technological Progress from China State Shipbuilding Corporation, and the Second Prize for Natural Science of Dalian City. He has accumulated years of field observation experience on island-reef geomorphology and wave hydrodynamics in the South China Sea.

Presentation Outline

This lecture examines the rising hazard potential of natural disasters driven by climate change and sea level rise, along with strategies for adapting to natural variability. It begins by exploring the linkages between climatic context and tipping points. The presentation then highlights key research findings on coastal erosion and its underlying mechanisms for Hainan Island and islands and reefs under climate change scenarios. Finally, the lecture analyzes urgent tasks for climate change

adaptation and calls for individual reflection and action toward mitigation of, and adaptation to, climate change.



Euda Helena Miranda

Presentation Title

Integrating Land Information for Better Governance: Cabo Verde's Reforms, Achievements and Lessons Learned

Speaker's BIO

Euda Helena Miranda is an Executive Board Member of the National Institute for Land Management of Cabo Verde (INGT), with extensive experience in land governance, cadastral systems, territorial planning, GIS, and the Cabo Verde Spatial Data Infrastructure (SDI). She has contributed to strategic initiatives modernizing land administration, integrating land information, and developing geospatial platforms for evidence-based decision-making. She currently oversees projects on land cadastre, land regularization, information-system interoperability, and territorial governance capacity, promoting more efficient, transparent, and sustainable land management in Cabo Verde.

Presentation Outline

This presentation shares Cabo Verde's experience in modernizing land administration through the integration of land information systems, highlighting the context of land governance, reforms to the land administration system, integration of land information, achievements and impacts, lessons learned, and future priorities; integrated land information serves as a strategic foundation for effective land governance, helping to secure land rights, support informed decision-making, improve public services, and promote sustainable territorial development.



Wenting CAO

Presentation Title

Satellite Remote Sensing for Integrated Land-Sea Governance in Pacific Island Countries

Speaker's BIO

Dr. Wenting CAO is an Associate Researcher at the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, MNR, China. She received her bachelor's and Ph.D. degrees in geography from Zhejiang University. Her research focuses on remote sensing monitoring of coastal and offshore environments, including time-series coastal change, mangroves, tidal flats, salt marshes and island ecosystems. She has led several research projects, including those funded by the National Natural Science Foundation of China, published over 20 academic papers, and participated in international cooperation on remote sensing monitoring and capacity building for Pacific Island Countries.

Presentation Outline

Pacific Island Countries and Territories (PICTs) are highly vulnerable to sea-level rise, tropical cyclones, coastal erosion, coral bleaching, and increasing development pressure. Based on work on remote sensing monitoring and capacity building for marine and coastal ecosystems in PICTs, this presentation introduces multi-source satellite applications for coastal erosion and expansion, benthic habitat and bathymetry mapping, coral bleaching and recovery, mangrove change, human settlement expansion and disaster monitoring. It further discusses how satellite remote sensing products, national reports, the PI-ReSOC platform, and

capacity-building activities can support integrated land-sea governance, ecological conservation, disaster response, and sustainable development in Pacific Island Countries.



Ming XIAO

Presentation Title

Integrating CFD-PINN Hybrid Modeling and Multi-Platform Observations for Early Warning of Tropical Harmful Algal Blooms: The Hainan Pilot Study

Speaker's BIO

Dr. Ming XIAO is an Associate Research Fellow at the Hainan Academy of Marine Economic Development and Resource Protection. He earned his PhD in Fluid Mechanics from École Centrale de Lyon in France. As a two-time Marie Skłodowska-Curie Fellow (EU), his expertise lies in marine hydrodynamics and AI-based environmental engineering.

Previously, he led research and development on coastal pollution in Spain for the EU. Currently, he focuses on reducing tropical red tides in Hainan by developing innovative forecasting methods that combine CFD with Attention-BiLSTM models and deploying IoT-enabled systems for dispersing modified clay.

Presentation Outline

The 2026 Hainan harmful algal bloom (HAB) crisis highlighted critical gaps in predicting bloom dynamics under tropical monsoon conditions. Here, we propose an integrated framework that combines high-resolution benthic cyst mapping, a 22-node stereo-observing array, and a novel hybrid model embedding locally derived temperature-growth kinetics into a computational fluid dynamics (CFD)-coupled Physics-Informed Neural Network (CFD-PINN) to deliver 72-hour forecasts of *Pheopolykrikos hartmannii* and *Margalefidinium* species. This study provides a mechanistically grounded, transferable template for AI-driven coastal hazard governance in warming tropical and subtropical seas.



Xiaoli DING

Presentation Title:

Spatial Information to Facilitate Sustainable Urban Development in Hong Kong

Speaker's BIO

Professor Xiaoli DING is Director of Research Institute for Land and Space and Chair Professor in the Department of Land Surveying and Geospatial Science at The Hong Kong Polytechnic University. He has taught at Northeastern University in China, Curtin University in Australia, and The Hong Kong Polytechnic University. His main research interests include land development, remote sensing, satellite positioning, intelligent sensing, disaster monitoring and management, and structural health monitoring. He has received a number of research awards.

Professor Ding is a Fellow of the Hong Kong Institute of Engineering Surveyors, a Fellow of the Institution of Civil Engineering Surveyors (UK), a Fellow of the Royal Institution of Chartered Surveyors, and a Fellow of the International Association of Geodesy.

Presentation Outline

Hong Kong is a high-density city that lacks land and space for development compared to many other cities in the world. It is therefore necessary to continuously optimize approaches to develop new land and make the best use of available land. To support such efforts, we have developed a spatial information platform that contains information such as topography, land cover and land use, land planning and zoning, distribution of existing building stocks, population and facilities (e.g., schools, hospitals and clinics, restaurants, shopping centers), transport systems (roads and railways), and walkways. We provide some case

studies of using spatial information in land use analysis including identifying the distribution of available land for development, studying the adequacy of supporting facilities in different districts of Hong Kong, and researching the delicate balance of development and preserving vegetated areas.



Tiejun WANG

Presentation Title

Leveraging GeoAI and Earth Observation for Large-scale Biodiversity Monitoring

Speaker's BIO

Tiejun WANG is a Professor of Geospatial AI for Biodiversity at the School of Geospatial Engineering and Science, Sun Yat-sen University, China. He is a recipient of China's national-level overseas high-level talent programme and a Distinguished Scholar under Sun Yat-sen University's Yat-sen Scholar Program. His research focuses on integrating Earth observation, remote sensing, GIS and artificial intelligence to address ecological and biodiversity challenges, with particular emphasis on developing scalable approaches for wildlife and biodiversity monitoring across large and remote landscapes. His research aims to advance GeoAI and Earth observation as scalable and operational tools for biodiversity monitoring and conservation decision-making.

Presentation Outline

This presentation will demonstrate how Earth observation and artificial intelligence can be integrated to detect, identify and count wildlife populations across extensive landscapes. Drawing on applications ranging from very-high-resolution satellite monitoring of African elephants and the great wildebeest migration in the Serengeti–Mara ecosystem to AI-assisted analysis of large-scale aerial wildlife surveys in the Boma–Badingilo landscape of South Sudan, I will illustrate how GeoAI can transform both satellite and aerial imagery into spatially explicit information on wildlife populations. The presentation will also discuss the opportunities and remaining challenges for moving from individual

research applications towards scalable and operational biodiversity monitoring systems that can support conservation and evidence-based decision-making.



Jie SHEN

Presentation Title

GeoAI-Enabled Actionable Cartography for Climate Resilience and Disaster Risk Management

Speaker's BIO

Jie SHEN has been a Professor of Cartography and Geographic Information Systems at the School of Geography, Nanjing Normal University (NJNU), since 2017. She is a recipient of the honorary title of **Jiangsu Provincial Distinguished Teacher**. She has been teaching cartography and GIS for over 25 years. Her main research interests include cartography for early warning and disaster management, navigation map design and generalization, mapping for the SDGs, and narrative mapping. She served as Co-chair of the ICA Commission on Cartography for Early Warning and Crisis Management (2023–2027); Secretary and Executive Member of UN-GGIM-AN (2025–2026); and a registered expert for ISO/TC 268/SC 1/WG 6 (Smart community infrastructures — Disaster risk reduction) since 2021. She has led two National Key Research and Development Program projects in cooperation with the Czech Republic and Japan, and four National Natural Science Foundation of China projects. She has published over 100 papers in academic journals and conference proceedings.

Presentation Outline

In response to the growing disaster risks posed by climate change in the Asia-Pacific region, this presentation explores how GeoAI-enabled, human-centred actionable cartography can transform multisource geospatial information into trustworthy, understandable and actionable knowledge to support risk identification, scenario analysis and evidence-informed decision-making. Drawing on research in dynamic urban flood

mapping, it demonstrates the potential of this approach to strengthen disaster risk management and climate resilience. The presentation concludes by proposing three actions for the Asia-Pacific region: developing common frameworks, advancing regional pilot initiatives and strengthening capacity development, thereby contributing to a more inclusive, resilient and sustainable future.



Chandrika Jagath Wellage

Presentation Title

From Geospatial Data to Climate Resilience: Sri Lanka's Experience in Integrated Geospatial Information and Disaster Response to Cyclone Ditwah

Speaker's BIO

C. J. Wellage is a senior professional in land surveying, geospatial information management, and land administration in Sri Lanka. He currently serves as a Senior Superintendent of Surveyors at the Surveyor General's Office and has extensive experience in cadastral surveying, geospatial data management, digital mapping, and land information systems. He has been actively involved in the modernization and digital transformation of Sri Lanka's land administration and surveying services, including digital cadastral workflows, GNSS-based surveying, GIS applications, and geospatial information systems. He also contributes to professional development, capacity building, and policy-related activities in the surveying profession. He has represented Sri Lanka at international forums, including UN-GGIM, and contributed to discussions on geospatial information, GeoAI, integrated land administration, climate resilience, and sustainable development.

Presentation Outline

The presentation will highlight Sri Lanka's experience and current challenges in managing geospatial information, particularly the fragmentation of important datasets across institutions and the need for an integrated and interoperable national geospatial information platform. Drawing on the experience of Cyclone Ditwah, it will demonstrate the importance of combining authoritative geospatial data, Earth observation, LiDAR, Digital Elevation Models, GIS analysis, and field verification for

effective disaster response and climate resilience. It will also highlight Sri Lanka's priority to strengthen its national geospatial data infrastructure in line with the UN-IGIF and progressively introduce GeoAI for rapid mapping, change detection, exposure and vulnerability analysis, and predictive decision support.



Mu YANG

Presentation Title

From Metacognition to Meta-Action: GeoMCP and Geospatial Information Public Goods for Climate-Resilient Decision-Making

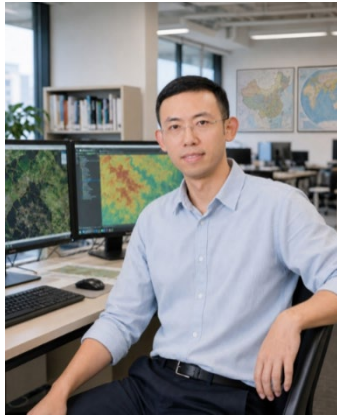
Speaker's BIO

Dr. YANG is a Hangzhou Overseas High-Level Talent. He previously served as Chief Data Scientist at IPG, and as a Project Researcher in the Department of Earth System Science at Tsinghua University. He currently serves as Distinguished Expert at the Institute of Global Geospatial Information and Sustainable Development, Moganshan Geospatial Information Laboratory, where he leads the development of the Global Geospatial Information Intelligence Framework and GeoMCP. His research focuses on Video GIS, spatial positioning, and spatiotemporal AI early-warning platforms. He has led multiple international and national-level projects and has received honors including the Best Practice Award in the China Region of the UN Big Data Hackathon.

Presentation Outline

Global climate change demands that data science shift from understanding climate to empowering resilience actions. Starting from metacognition, this report proposes a "meta-action" framework and re-examines scales, dimensions, and beyond. Using flood risk assessment as

a case, it explains how GeoMCP protocolizes geospatial analysis tools, enabling AI agents to orchestrate spatial decisions within trusted boundaries. Geospatial information public goods provide open, reusable data and institutional foundations. Together, they transform geospatial knowledge into verifiable, scalable, and equitable climate-resilient actions for sustainable development.



Yinxi GONG

Presentation Title

Unlocking the Value of Urban Territorial Spatial Monitoring Data for Digital and Intelligent Urban Spatial Governance

Speaker's BIO

Dr. Yinxi GONG is a professor-level senior engineer and Deputy Director of the Department of Geospatial Information Analysis at the National Geomatics Center of China. His work focuses on geospatial analysis, spatiotemporal data mining, AI-enabled GIS, and monitoring and evaluation of territorial spatial planning implementation. He has led or managed multiple national and provincial/ministerial-level research and engineering projects. The urban territorial-space monitoring models and tools developed under his leadership have been integrated into the Ministry of Natural Resources' national territorial spatial planning "One Map" system, supporting routine monitoring in 297 cities. He has published over 30 papers, holds more than 10 granted invention patents, contributed to four industry standards, and his work received the First Prize of the National Outstanding Territorial Spatial Planning Award.

Presentation Outline

This presentation explores how urban territorial spatial monitoring data can support digital and intelligent urban spatial governance. It introduces the programme's background, data system and practical outcomes, focusing on the monitoring and analytical framework covering all prefecture-level and larger cities in China. Built on high-quality datasets, geographic units, a model library and multidimensional visualization, the framework links data governance, analytical modelling and decision support. Case studies illustrate applications in planning implementation oversight, urban

agglomeration and metropolitan-area analysis, equal access to public services, revitalization of underused urban space, and safety and resilience assessment. Looking ahead, artificial intelligence will be integrated across monitoring, analysis, evaluation and application to unlock the value of data, support people-centred urban development, and advance citywide digital transformation.



Binru ZHAO

Presentation Title

Intelligent Interpretation of Remote Sensing Enables Sustainable Coastal Zone Management

Speaker's BIO

Binru ZHAO, Engineer of the National Marine Data and Information Service (NMDIS), has long been dedicated to intelligent interpretation of coastal remote sensing data, focusing on building high-quality sample datasets and conducting research on remote sensing identification and dynamic monitoring of typical coastal zone features.

Presentation Outline

Research Progress and Applications of Intelligent Interpretation in Coastal Zone Remote Sensing: This specifically includes the construction of high-quality remote sensing sample sets for coastal zones, intelligent interpretation methods for typical coastal elements, dynamic monitoring application services, as well as comparative analyses of various visual models and task-specific model selection methods.



Mao ZHU

Presentation Title

SAR-AI Fusion Technology: Exploring New Pathways for Intelligent Remote Sensing Monitoring

Speaker's BIO

Mao ZHU has been engaged in research on spaceborne SAR/InSAR algorithms and their applications. In 2018, he assumed the role of Chief Technology Officer at Beijing Orients Technology Co., Ltd., where he has been dedicated to advancing the engineering applications of SAR technology. In the field of urban construction, he led the development of a city-scale building deformation risk survey system, which has been deployed in over 50 cities across China. In the domain of geohazards, he conducted research on the Baige landslide and landslide risks in various provinces of Southwest China. In the water resources sector, he carried out deformation monitoring of major dams including Xiluodu and Baihetan, as well as medium- and small-sized reservoirs. In the field of natural resources, he integrated machine learning approaches with SAR data for cultivated land classification, supporting the monitoring of non-agricultural and non-grain use of farmland. He has authored 5 SCI-indexed papers and holds over 10 authorized invention patents.

Presentation Outline

The all-weather, day-and-night observation capability of spaceborne SAR, combined with the rapid advancement of artificial intelligence, has opened up new possibilities for the intelligent transformation of remote

sensing monitoring. This presentation provides a systematic review of the technical pathways and current state of SAR-AI integration, with a focus on the breakthroughs enabled by deep learning in SAR image target detection, land cover classification, and deformation field modeling. Building on this foundation, the presentation examines key challenges including multi-source data fusion, generalization capacity of large-scale models, and the incorporation of physical constraints. Through typical application scenarios such as urban infrastructure safety and geohazard identification, the presentation further explores the prospect of SAR-AI integration evolving from "intelligent interpretation" to "intelligent cognition."



Xin ZHAO

Presentation Title

Geo-AI Enabled Inspection Solutions to Boost Climate Resilience Across the Asia-Pacific

Speaker's BIO

Mr. Xin ZHAO is an experienced Business Development expert focused on geospatial sensing, 3D laser scanning and global commercial development of spatial information technology. With more than 11 years of industry experience, he has participated in overseas market exploration and international business cooperation across multiple continents. At present, he works as Business Development Director of South Survey. He concentrates on strategic partner cultivation, overseas market expansion and the commercial implementation of LIDAR and Geo-AI solutions for infrastructure, environmental monitoring and digital twin scenarios.

Presentation Outline

This presentation introduces South Surveying's self-developed air-ground integrated intelligent inspection system. Supported by the South Differential Augmentation Service (SDAS), the solution integrates multi-sensing quadruped robots equipped with GNSS, LiDAR SLAM, cameras and environmental sensors, and is centrally orchestrated by the SkyMirror low-altitude economy platform to strengthen climate disaster resilience across the Asia-Pacific.



Meng LI

Presentation Title

Geospatial Intelligence Helps Creating a Smarter World

Speaker's BIO

Mr. LI has extensive expertise in 3D GIS technologies, software design, and real-world applications. He focuses on advancing the research and development of next-generation 3D GIS.

Presentation Outline

Geospatial intelligence is the core foundational infrastructure for building a smarter, more sustainable world. SuperMap's geospatial AI technologies empower digital twin construction and smart city development worldwide. These technologies are widely applied in urban management, natural resource governance, energy systems, and climate adaptation. By converging 3D, AI, and distributed GIS, SuperMap transforms massive geodata into actionable insights, fostering evidence-based planning and operational efficiency. This intelligence accelerates global digital transformation, making cities safer, greener, and more responsive. Overall, SuperMap empowers both public and private sectors to proactively tackle challenges, driving long-term sustainability and helping organizations optimize daily operations and formulate long-range strategies.



Chisheng WANG

Presentation Title

Relative Sea-Level Rise in Coastal Cities: Land Subsidence Monitoring, Risk Assessment, and Adaptation Responses

Speaker's BIO

Professor Chisheng Wang is a Distinguished Professor and Doctoral Supervisor at Shenzhen University. His research focuses on satellite geodesy, deformation monitoring of natural hazards and infrastructure, and sustainable development of coastal cities.

Presentation Outline

Relative sea-level rise is jointly driven by oceanic change and land subsidence, intensifying flooding, erosion, and asset exposure in coastal cities. This report integrates InSAR-based subsidence monitoring, coastal exposure assessment in Europe, and global analyses of urban adaptation to examine the underlying mechanisms, spatial disparities, and societal impacts. The results highlight pronounced subsidence in Asian deltaic cities, the substantial influence of vertical land motion on future coastal risk, and the limited spatial response of most coastal cities to rising sea levels.



Baogui KE

Presentation Title

Height Datum Transfer and Unification for Islands and Reefs

Speaker's BIO

Baogui KE is a Research Professor at the Chinese Academy of Surveying and Mapping (CASM). His research centers on the theory and applications of satellite altimetry, with particular emphasis on marine gravity fields and depth datums. He was a key contributor to data processing for the new-generation National Gravity Reference Network and developed a regional gravity field model for the South China Sea, which has served as a ground validation reference for multiple Earth observation satellite datasets. He has led more than ten national and provincial/ministerial research projects and has authored over twenty papers, twelve of which are SCI/EI-indexed. He has been granted nine invention patents and seven software copyrights, and has received three provincial/ministerial awards. He also serves as a master's supervisor at several universities, a committee member of academic societies, and a reviewer for academic journals.

Presentation Outline

The unification of vertical datums between islands and the mainland is crucial for the development of geospatial information products under a consistent spatial reference framework. In this study, we utilize shipborne gravity data, satellite altimetry, global geopotential models (GGMs), GNSS observations, and tide gauge records to determine the geopotential difference between two points on the mainland and an island. Based on the theory of potential and the principles of astronomical leveling, we implement an iterative geopotential-difference approach to realize height datum transfer across the sea. The results of this research are of significant importance for the establishment and unification of vertical

datums in island nations and regions.



Chao DONG

Presentation Title

Advances and Prospects in Marine Unmanned Swarm Detection Technology

Speaker's BIO

Chao Dong is a Research Professor. He serves as Director of the Key Laboratory of Marine Environmental Survey Technology and Application, Ministry of Natural Resources, and Chief Scientist of the Marine Unmanned Intelligent Equipment Team at the Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai). He has long been dedicated to the research and development of marine unmanned intelligent equipment and key technologies for networked observation and detection.

Presentation Outline

This talk presents research progress on USVs, USV swarms, USV-UAV coordination, and air-sea-underwater collaboration for marine environmental surveys, together with an outlook on future directions for marine unmanned swarm detection.



Nan XU

Presentation Title

Tidal Flat Topography Inversion from Multi-source Satellite Altimetry and Remote Sensing Imagery

Speaker's BIO

Nan Xu is an Associate Professor at Shenzhen University, with research interests in coastal zone remote sensing. In recent years, as first/corresponding author, he has published over 60 papers in journals including Nature Communications, PNAS, Science Bulletin, Remote Sensing of Environment, and Geophysical Research Letters. He has been repeatedly listed in the World's Top 2% Scientists (annual impact). He has served as Principal Investigator for projects funded by the National Natural Science Foundation of China (Young Scientist Program and General Program) and the Jiangsu Provincial Natural Science Foundation (General Program). He is an Associate Editor for All Earth and The Egyptian Journal of Remote Sensing and Space Sciences, and serves on the Editorial Board of International Journal of Digital Earth and Science of Remote Sensing.

Presentation Outline

Tidal flats, as a critical zone of land-sea interaction, play a key role in defending against storm surges, mitigating wave erosion, and maintaining coastal ecosystems. This report consists of two parts: first, constructing a large-scale, high-precision terrain inversion framework for China's tidal flats based on the SWOT satellite; second, integrating ICESat-2 and long-term multi-source satellite imagery to continuously track the interannual variations in the topography of Jiangsu's tidal flats. This study provides data support and a scientific basis for coastal disaster prevention and mitigation and for sustainable development.



Chunxiao WANG

Presentation Title

Exploration and Application of Tropical Remote Sensing Intelligent Interpretation Technology: A Case Study of Hainan Island

Speaker's BIO

Chunxiao WANG received his Ph.D. from Shandong University of Science and Technology in 2015. He currently serves as a professor of engineering at Hainan Geomatics Center, Ministry of Natural Resources of China. His research interests include intelligent remote sensing interpretation and intelligent natural resource management.

Presentation Outline

This presentation mainly introduces the software achievements, sample library, models, and application progress of intelligent remote sensing interpretation for tropical features on Hainan Island. We have completed approximately 500,000 samples for intelligent remote sensing interpretation, covering distinctive tropical features such as buildings, cultivated land, forest land, rice, and rubber. A high-precision building interpretation model with an accuracy of 90% has been developed, which supports the construction of Hainan Province's 'One Map' geographic base map. Current ongoing work includes optimizing interpretation models for rice and aquaculture ponds, and developing intelligent agents.



Chen LIU

Presentation Title

Refined Tropospheric ZWD Model for the Tibetan Plateau Under Oceanic Moisture Influence

Speaker's BIO

Chen LIU received a Ph.D. degree in Geodesy and Surveying Engineering from Wuhan University, China, in 2023 and is a postdoctoral researcher at National Geomatics Center of China. His research interests include high-precision GNSS data processing, tropospheric inversion, 3D water vapor tomography.

Presentation Outline

Oceanic moisture transport modulated by complex terrain causes pronounced spatial heterogeneity of tropospheric water vapor over the Tibetan Plateau, challenging high-precision GNSS positioning and water vapor monitoring. Traditional models such as the Global Pressure and Temperature 3 (GPT3) model suffer from limited spatiotemporal resolution and inadequate vertical correction, exhibiting suboptimal performance over the plateau. This study develops a refined Tibetan zenith wet delay (ZWD) model (TZ) based on 0.25° European Centre for Medium-Range Weather Forecasts Reanalysis 5 (ERA5) data, adopting a cubic polynomial framework combined with semidiurnal harmonic calibration. The model effectively captures spatiotemporal variations of tropospheric water vapor and enables accurate ZWD estimation at arbitrary times and altitudes. Validated against ERA5 and radiosonde data, the TZ model outperforms GPT3 in bias and RMSE at all altitude levels, delivering reliable ZWD estimates across the entire plateau.



Yi DENG

Presentation Title

Integrated RS and GIS in Coastal Eco-Geological Surveying: Insights from Representative Case Studies

Speaker's BIO

Yi DENG joined Guangzhou Marine Geological Survey (GMGS) in 2013. He rotated through the following departments: the Institute of Marine Technical Methods, the Technical Center of Natural Resources Supervision, and the Institute of Marine Ecological Environment. He is dedicated to Marine Navigation, RS Interpretation and Data Comparison. He also has extensive practical experience in both marine and terrestrial surveying, as well as geo-data processing.

Presentation Outline

The presentation will share some practical experiences in Coastal Eco-Geological Surveying, using certain case studies as inspiration. This presentation focuses on how vital GIS can be in turning fragments from RS Interpretation into time-space relevant evidence, so as to support studies on how coastlines changed and why they did.



Ziyan ZHAO

Presentation Title

Integrated Space-Air-Ground-Sea Digital Twin System: Empowering Coastal Resilience Building and Smart Ocean Monitoring

Speaker's BIO

Ziyan Zhao is the General Manager of Feiyan Remote Sensing, the leading Chinese private aerial photography enterprise. Frank holds a Master's degree from Boston University. He steered the company's transition from traditional surveying to a data-and-application service provider. He has received the Second Prize of the MOE Outstanding Scientific Research Achievement Award, the Third Prize of the Hubei Provincial Science and Technology Progress Award, six national and five provincial outstanding engineering awards, and he also holds eight invention patents. He has been named a Wuchang Talent, a Forbes China 30 Under 30 honoree, and an advanced individual in the provincial surveying and mapping industry.

Presentation Outline

This presentation centers on a digital twin ocean platform, showcasing the practical applications of integrated space-air-ground-sea monitoring technologies in marine patrol, submarine cable inspection, and dynamic sensing of coastal zones. By fusing multi-source data with intelligent interpretation, we construct dynamic digital twin models that support disaster early warning, ecological assessment, and resilience planning decisions. Grounded in the United Nations Sustainable Development Goals, we share our experience in data sharing and international cooperation, aiming to provide replicable solutions for ocean monitoring and digital twin applications to coastal regions worldwide.



Tang XIANG

Presentation Title

Practice of Land Sea Integrated 3D Visualization Technology and Reflections on Its Application Potential for Island Countries

Speaker's BIO

Mr. Xiang Tang, Senior Engineer and Registered Surveyor, is the Team Leader of the Marine Surveying and 3D Visualization Application Service Team at the Sanya South China Sea Geological Research Institute, Guangzhou Marine Geological Survey. He also serves as an adjunct Master's Supervisor at the College of Marine Technology and Equipment, Hainan University. His research interests focus on integrated visualization and application of multi-source big data from land and sea. He has extensive hands-on technical experience and proven expertise in project organization and implementation, particularly in the development of the Tianditu (Map World) platform and marine thematic application systems.

Presentation Outline

Land-sea spatial integration serves as a critical pillar for marine governance and sustainable development in Asia-Pacific island countries. This report focuses on the integrated 3D visualization of islands and reefs across land and sea, carrying out the integration of multi-source geographic information data and the practice of 3D visualization technology, with the aim of constructing a reusable 3D spatial governance solution for island and reef coastal zones. In light of the current status of geographic information infrastructure in Asia-Pacific island countries, the report explores the application value of this technology in scenarios such as the development and utilization of island and reef coastal zones, marine ecological protection, and marine disaster prevention and mitigation, thereby providing a reference for regional

marine spatial governance.

Dheny Trie Wahyu Sampurno
Presentation Title

Speaker's BIO

Presentation Outline

Okieno Samuels

Presentation Title

Building Coastal Resilience through Integrated Geospatial Information:
Jamaica's Experience

Speaker's BIO

Presentation Outline



Huaqiao XING

Presentation Title

Natural resources intelligent monitoring and knowledge services

Speaker's BIO

HuaqiaoXING, Vice Dean of the School of Surveying, Mapping, and Geospatial Information at Shandong Jianzhu University. His research primarily focuses on natural resource monitoring and geospatial information service computing. He has led over 20 vertical and horizontal projects, including those funded by the National Natural Science Foundation of China, published more than 100 papers, applied for over 10 national invention patents, and received over 10 provincial and ministerial-level awards, including the Second Prize of the Ministry of Natural Resources Science and Technology Progress Award and the First Prize of the China Surveying and Mapping Science and Technology Progress Award.

Presentation Outline

The rapid rise of low-altitude economy and Earth observation brings both opportunities and challenges to natural resources monitoring. Despite abundant multi-source data, bottlenecks persist: inaccurate change detection, inflexible workflows, and underutilized results. This report addresses these gaps via a systematic framework—Precise Monitoring, Intelligent Computing, Knowledge Services. Our contributions include: an adaptive change detection framework with multi-feature fusion for high accuracy; a "share-reuse-integrate" service model for overcoming interoperability barriers; and the self-developed ILCS platform synergizing low-altitude and satellite remote sensing for integrated monitoring and knowledge applications. This work holds theoretical value and practical significance, aligning with national strategies and regional innovation priorities.



Yunqiang ZHU

Presentation Title

Automated Geo-computations Driven by Knowledge Graphs

Speaker's BIO

The dean and professor of the School of Aerospace Intelligence, Hainan University. His main research fields include data sharing in earth science, geographic spatial data ontology and knowledge graphs. He has published 8 monographs and over 200 papers, obtained 9 national invention patents, participated in the formulation and release of 5 national standards, led the formulation of 4 industry standards. He also won the second prize of the National Science and Technology Progress Award.

Presentation Outline

Geo-computation is a process of applying geographic models to geographic data, which reveals the distribution patterns of geographic entities, events and phenomena, simulates their evolution processes, and predicts their development trends. It is a key method in geographic information science. Geo-computation is transitioning from the independent application stage and distributed methods to the cloud computing stage, and now is developing towards the direction of automation and intelligence. This study proposes a knowledge graph-driven approach for automating geo-computations. Based on the constructed geo-computation knowledge graph, an algorithm for model matching and linking is developed, and a set of knowledge graph-driven automated computing processes is designed. To apply and validate the knowledge graph-driven approach, geomorphological classification, soil erosion, and soil production potential are selected as case studies.



Hui JIANG

Presentation Title

Deepening International Cooperation on Geospatial Information for Joint Climate Resilience and Sustainable Development in the Asia-Pacific

Speaker's BIO

Dr. Hui JIANG, Senior Engineer and Doctor of Economics, has served as Secretary-General of the Asia-Pacific Space Cooperation Organization (APSCO) since February 2025. Previously, she was Counsellor at the Embassy of the People's Republic of China in the Russian Federation and China's Space Representative to Russia, and Division Director at the Department of International Cooperation, China National Space Administration (CNSA).

Dr. JIANG has extensive experience in international space cooperation. She has advanced bilateral space collaboration with multiple countries and coordinated major multilateral projects, including the BRICS Remote Sensing Satellite Constellation and the International Lunar Research Station (ILRS). Under UNCOPUOS, she led China's participation in the consultations on the Guidelines for the Long-Term Sustainability of Outer Space Activities (LTS), and served as Co-Chair of the Action Team on Global Partnership in Space Exploration and Innovation under UNISPACE+50. She also has rich experience in space product import and export management and in promoting domestic space-related regulations and mechanisms.

Presentation Outline

This keynote speech centers on deepening international cooperation on geospatial information for joint climate resilience and sustainable development in the Asia-Pacific. Set against the global climate

governance agenda and the UN Integrated Geospatial Information Framework (UN-IGIF), it articulates APSCO's role as a bridge between global frameworks and national realities, showcases the organization's operational practice built on the Data Sharing Service Platform (DSSP) and full-spectrum capacity building programs, explains the core values of regional intergovernmental collaboration, and outlines four priority initiatives for advancing multi-stakeholder geospatial partnership across the region.



Shujuan JIA

Presentation Title

Exploration of S 100 based Nautical Chart Production, Quality verification and Application

Speaker's BIO

Shujuan JIA, Senior Engineer, is Deputy Section Chief of the Cartography Division, Shanghai Chart Center. With nearly 20 years of experience in nautical chart production and application, she focuses on S-100 standards development and production technologies. She led the development of the full-process production system for China MSA's new-generation S-101 ENC's, enabling large-scale production of S-101 ENC's for China's coastal waters. She has extensive experience in S-101 ENC production and application.

Presentation Outline

Based on the International Hydrographic Organization (IHO) S-100 Universal Hydrographic Data Model, this presentation introduces the technical strengths and product system of the new-generation nautical chart standards, and summarizes the research progress and supporting software development of China Maritime Safety Administration for the S-100 standards. It mainly describes the complete production process of S-100 series products supported by the Cangqiong Nautical Chart Production System, including data import, editing and processing, quality verification, and final product packaging.



Hongping ZHANG

Presentation Title

National Platform for Common Geospatial Information Services
(Tianditu)

Speaker's BIO

Hongping ZHANG, Senior Engineer at the National Geomatics Center of China. As the technical lead, he has long been engaged in the development and application of the National Platform for Common Geospatial Information Services. He currently serves as Co-Chair of the Working Group on Geospatial Information for Disaster Risk Management (WG GI4DRM) under the United Nations Committee of Experts on Global Geospatial Information Management (UN-GGIM), and as Co-Chair of the Working Group on Spatial Data Engineering, Databases and Scalable Data Infrastructures under ISPRS Technical Commission IV (ISPRS TC IV).

Presentation Outline

China has been steadily advancing the development of an integrated geospatial information framework. In 2011, it officially launched the National Platform for Common Geospatial Information Services (Tianditu)—to promote the sharing, integration, and efficient utilization of government geospatial resources, thereby empowering the digital economy and supporting the construction of Digital China. This presentation systematically reviews the platform's development background, strategic objectives, and overall architecture, summarizes its application service outcomes, and explores future development directions, with the aim of providing useful references for geospatial information sharing governance

and spatial digital infrastructure development in other countries.

Nidkombu Edwin
Presentation Title

Speaker's BIO

Presentation Outline

Hayato FUKUOKA
Presentation Title

Speaker's BIO

Presentation Outline